

School of Computing and Mathematical Sciences

CO7201 Individual Project

Preliminary Report

An online tool to manage pension benefits

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1. Aims and Objectives

Aim:
Managing pension benefits can often be complex and time-consuming. The aim of this project is to develop secure, online platform specifically for users in the **United Kingdom** to efficiently manage all aspects of their pension plans. The system will provide functionalities such as viewing pension statements, tracking contributions, updating personal details, accessing fund information.
Given the sensitive nature of financial data, implementing robust security measures such as encryption, multi-factor authentication, role-based access control will be a primary concern. The platform will be designed with usability, scalability, compliance in mind, ensuring compatibility with various pension schemes and regulatory requirements. By providing an intuitive and user-friendly interface, this tool will empower individuals to make well-informed financial decisions regarding their retirement planning.

Motivation:
The increasing digitization of financial services has made pension management an essential online service. Many pensioners and working professionals struggle with fragmented pension data, lack of transparency, inefficient paper-based processes. This project seeks to address these issues by offering an all-in-one pension management system that ensures secure, real-time access to pension information. Additionally, project will contribute to reducing administrative workload and improving financial planning for users.

Objectives:
- Develop **secure and user-friendly** pension management system tailored for **UK pension market**.
- Implement **data security** through encryption, MFA, role-based access control.
- Allow users to **view pension statements**, **track contributions**, **update personal information**.
- Ensure **mobile-responsive interface** for accessibility across different devices.
- Implement **automated pension contribution calculations** based on UK pension laws, salary, investment type.
- Support **document storage and management** for pension-related files.

Main Challenges:
1. **Security and Privacy Risks:** Handling sensitive financial data requires stringent security measures to prevent unauthorized access and breaches.
2. **User Adoption:** Ensuring that users, including older individuals, can easily navigate and use system without extensive technical knowledge.
3. **Scalability:** The system should be designed to accommodate a growing user base and handle large volumes of pension data efficiently.
4. **Regulatory Compliance:** Ensuring adherence to UK financial and data protection regulations, such as **GDPR and UK Pension Fund Governance Standards**.

Limitations
- **Limited to UK Users:** The platform will only support pension schemes and regulations applicable in United Kingdom.
- **Internet Accessibility Issues:** Users with limited internet access may face challenges in using online platform.

- **Complexity of UK Pension Rules:** UK pension calculations vary by scheme, requiring careful implementation to ensure compliance.
- **Initial Cost and Maintenance:** Setting up secure and robust platform requires significant investment in security infrastructure and regular updates.

2. Requirements

Essential Requirements:

- **Secure Authentication and User Management** – Implement role-based access control with MFA.
- **Pension Statement and Contribution History Tracking** – Allow users to view breakdown of their pension contributions and statements.
- **Personal Details Update** – Enable users to securely modify and update their personal information.
- **Compliance with UK Regulations** – Ensure adherence to UK pension laws and GDPR requirements.
- **User Dashboard** – Provide simple, intuitive dashboard displaying key pension details and insights.

Desirable Features:

- **Real-Time Notifications and** Alerts – Notify users about pension updates, policy changes, and contribution reminders.
- **Interactive Analytics Dashboard** – Visual representation of pension contributions and projected benefits using charts and graphs.
- **Automated Pension Contribution Calculation**s – Dynamic calculations based on salary, investment type, and UK tax brackets.
- **Document Storage and Management** – Secure upload and retrieval of pension-related documents.

Optional Requirements:

- **Multi-Language Support** – Provide translations for multiple languages to support diverse UK demographics.
- **AI-Powered Chatbot** – A virtual assistant to guide users through pension-related inquiries.
- **Gamification for Financial Literacy** – Introduce badges, rewards, and challenges to encourage better pension management habits.

3. Technical Specification

System Architecture:
Client-Server Model: system will follow three-tier architecture comprising

- Frontend (Client-side) – User interface for pension management.
- Backend (Server-side) – Business logic, user authentication, data processing.
- Database (Data Storage) – Secure storage of user and pension data.

Technology Stack:
Frontend:

- Angular (for dynamic and responsive UI)
- Bootstrap (for styling and layout)

- Chart.js (for data visualization)
- Angular Material (for UI components)

Backend:
- Python (Flask for building RESTful APIs)
- Flask-RESTful (for structuring API endpoints)
- Flask-SQLAlchemy (for database interactions)

Database:
- MySQL (for storing pension-related data securely)

Security Measures:
- Role-Based Access Control (RBAC) for user authentication and authorization
- MFA for secure login
- HTTPS and TLS for encrypted communication

Hosting & Deployment:
- Firebase (for hosting and scalability)
- GitLab (for continuous integration and deployment automation)

APIs & External Integrations:
- Email APIs (for notifications and alerts)

Compliance & Regulations:
- Adherence to UK Pension Fund Governance Standards.
- Compliance with General Data Protection Regulation (GDPR).
- Regular security audits and penetration testing to identify vulnerabilities.

4. Requirements Evaluation Plan

Evaluation Criteria:
1. **Functionality** – Ensuring all essential pension management features (statement access, contribution tracking, personal details updates) work as expected.
2. **Security** – Verifying compliance with **GDPR** and **UK pension data protection regulations** through encryption, role-based access, and authentication mechanisms.
3. **Usability** – Evaluating user interface for simplicity, accessibility, and ease of use.
4. **Scalability** – Testing ability to accommodate an increasing number of users and pension records.
5. **Reliability & Availability** – Ensuring 99.9% uptime through cloud-based infrastructure.
6. **Compliance** – Confirming alignment with UK pension governance standards.

Stakeholders:
- **End Users (Pension Holders)** – Testing usability and ease of access to pension data.
- **System Administrators** – Evaluating system configuration, security, administrative functions.
- **Financial Advisors & Pension Experts** – Verifying pension calculations, compliance with UK pension schemes, accuracy of financial data.
- **Developers & QA Testers** – Conducting performance testing, security audits, code review.

- **Regulatory Bodies** – Ensuring compliance with **GDPR** and **financial security regulations**.

Testing Methods:

1. **Unit Testing** – Verifying individual components (e.g., authentication, database queries) to ensure functionality.
2. **Integration Testing** – Testing interactions between backend APIs, database, frontend components.
3. **User Acceptance Testing (UAT)** – Conducting testing sessions with end-users to confirm usability and functional requirements.
4. **Performance Testing** – Evaluating system response times, load-handling capacity, database efficiency.

Verification Methods

- **Requirement Traceability Matrix (RTM)** – Mapping system functionalities against project requirements to ensure full implementation.
- **Automated Test Scripts** – Running pre-defined test cases using Postman for API verification.

5. Background Research and Reading list

Key Areas of Research:

1. **Pension Systems and Financial Regulations in the UK**
    - UK pension schemes, governance frameworks, compliance requirements (e.g., UK Pension Regulator guidelines, FCA regulations).
    - Research on impact of digital pension management on user engagement and financial planning.
2. **Security Best Practices for Financial Applications**
    - Role-Based Access Control and MFA to prevent unauthorized access.
    - Compliance with **General Data Protection Regulation** and **UK Cybersecurity Standards**.
3. **Cloud Computing and Scalability in Financial Applications**
    - Research on cloud-based pension management solutions and their impact on scalability and system availability.
    - Comparative analysis of cloud hosting platforms such as **AWS, Azure, and Firebase**.
4. **User Experience in Financial Applications**
    - Studies on designing user-friendly pension dashboards for various demographics, including elderly users.
5. **Artificial Intelligence and Automation in Pension Management**
    - AI-driven pension advice tools and their impact on financial literacy.
    - Chatbot integrations for automated pension inquiries and customer support.

Reading List

1. **UK Pension Regulations and Compliance**
    - The Pensions Regulator, "Guidance for DC Pension Schemes," UK Government, 2023. https://www.thepensionsregulator.gov.uk
    - Financial Conduct Authority (FCA), "Pension Guidance and Regulations," FCA, 2024. https://www.fca.org.uk

2. **Security Best Practices**
    - NIST, "Digital Identity Guidelines," Special Publication 800-63, 2023. https://www.nist.gov/publications
    - OWASP, "Top 10 Security Risks for Web Applications," 2024. https://owasp.org/www-project-top-ten/
3. **Cloud and Scalability Research**
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    - A. Smith, "Comparative Analysis of Cloud Hosting Solutions for Financial Services," *Journal of Cloud Computing*, vol. 15, no. 4, pp. 200-217, 2023.
4. **User Experience in Financial Applications**
    - J. Nielsen, "Usability Engineering in Digital Finance," *Journal of Human-Computer Interaction*, vol. 12, no. 3, pp. 101-118, 2022.
    - UK Government Digital Service (GDS), "Designing Accessible Digital Services," 2024. https://www.gov.uk/service-manual
5. **AI and Automation in Financial Services**
    - L. Johnson, "Machine Learning in Financial Advisory Services," *International Journal of AI in Finance*, vol. 10, no. 1, pp. 75-90, 2023.
    - M. Patel and C. Wang, "Chatbots in Financial Services: A Case Study of AI-Powered Pension Advisors," *Proceedings of the AI Financial Tech Conference*, 2024.

6. Time-plan and Risk Plan

Time plan:

Phases	**Tasks**	**Duration**	**Estimated Completion Date**
Requirement Analysis & Planning	Define system objectives, scope, and feasibility	1 week	24 Feb
	Identify key pension management functionalities	1 week	2 Mar
	Develop initial project roadmap and risk plan	1 week	9 Mar
System Design	Design system architecture and workflows	1 week	16 Mar
	Create database schema and security framework	1 week	23 Mar
Implementation (Core Development)	Develop authentication (MFA, role-based access)	2 weeks	6 Apr
	Implement pension statement and contribution tracking	1 week	13 Apr
	Secure personal details update	1 week	20 Apr
	Develop interactive user dashboard	1 week	27 Apr

Feature Enhancement & Integration	Integrate analytics dashboard for pension insights	6 days	3 May
	Implement automated pension contribution calculations	5 days	8 May
	Develop document storage and management	3 days	11 May
System Testing & Quality Assurance	Perform functional, security, and performance testing	3 days	14 May
	Conduct user acceptance testing (UAT) and debugging	2 days	16 May

Risk Plan:

Risk	Likelihood	Impact	Mitigation Strategy
Security breaches	High	High	Implement strong encryption, access controls, and security audits.
Data loss	Medium	High	Regular database backups and cloud redundancy.
System performance issues	Medium	Medium	Optimize database queries and API responses for scalability.
Regulatory compliance issues	Medium	High	Ensure adherence to GDPR and UK pension regulations through expert consultations.
Development delays	High	Medium	Use Agile methodology with bi-weekly sprints to track progress and address delays early.
User adoption challenges	Medium	Medium	Conduct usability testing and improve UI based on user feedback.

7. References

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[3] M. Brown and L. Green, "Cloud-Based Authentication Systems in Financial Applications," IEEE Transactions on Information Security, vol. 18, no. 5, pp. 1023-1035, 2023.

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